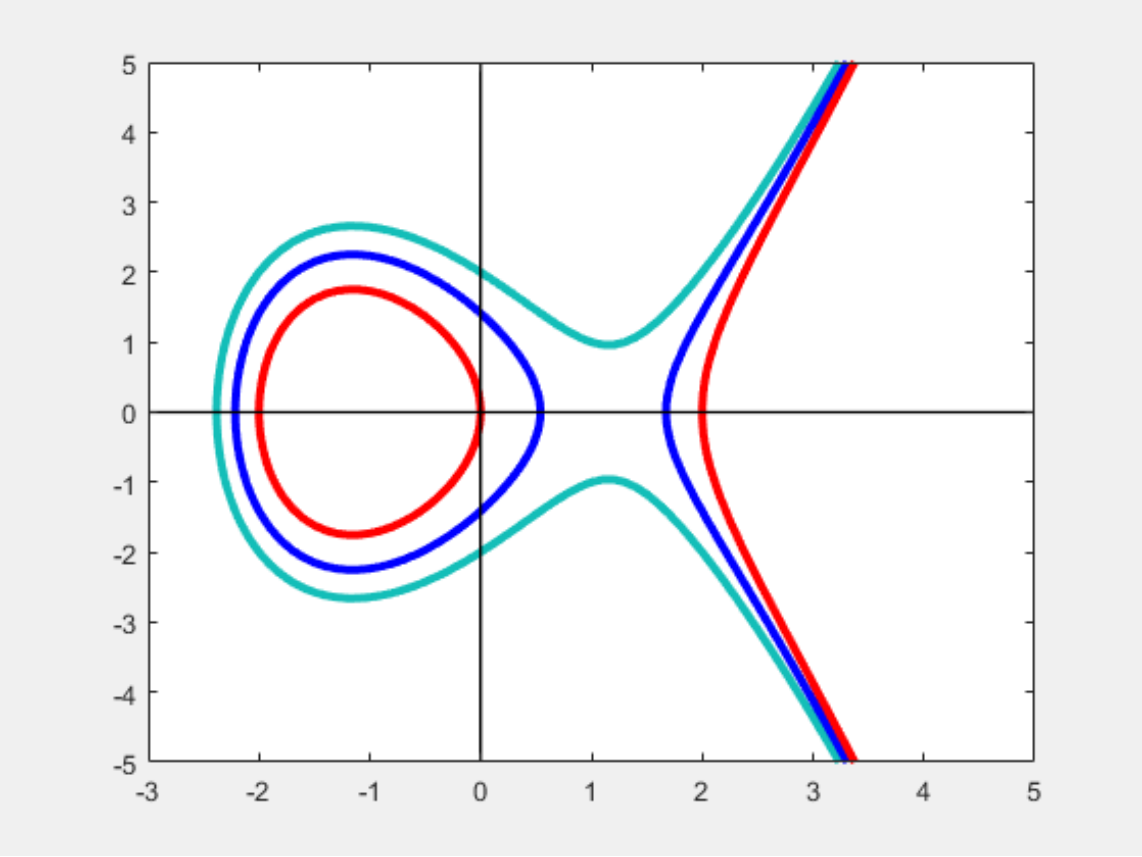
Introduction to Elliptic Curves
&
Their Applications

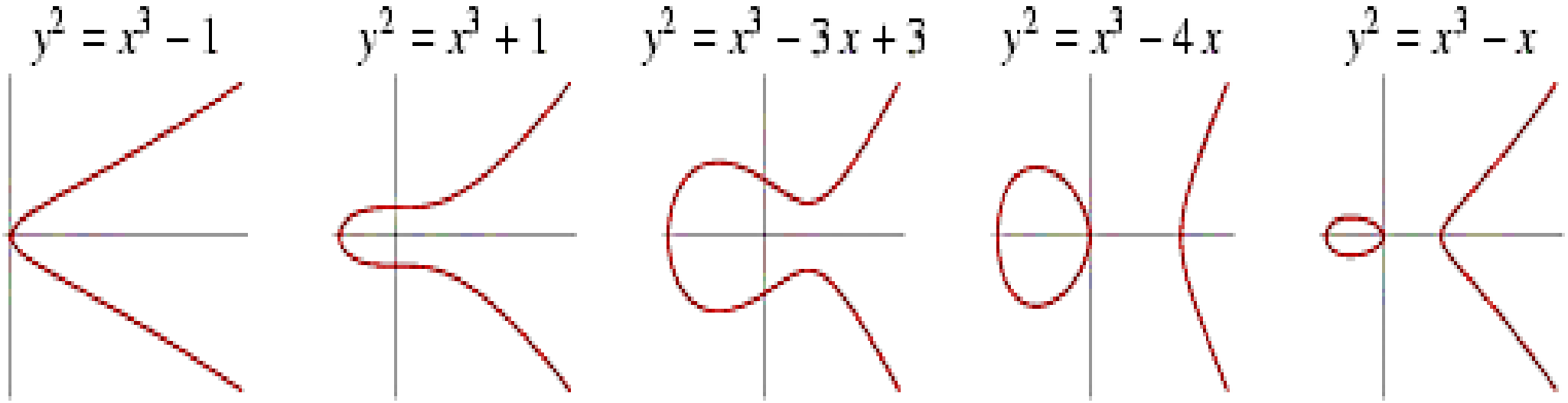


The general form of an Elliptic Curve

An ***elliptic curve*** is a plane curve defined by an equation of the form

$$\boxed{y^2 = x^3 + ax + b}$$

Examples



Characteristics of Elliptic Curve

- **What are Elliptic Curves?**
  - *Curve with standard form* $y^2 = x^3 + ax + b \quad a, b \in \mathbb{R}$

- **Characteristics of Elliptic Curve**
  - *Forms an abelian group*
  - *Symmetric about the x-axis*
  - *Point at Infinity acting as the identity element*

Characteristics of Elliptic Curve

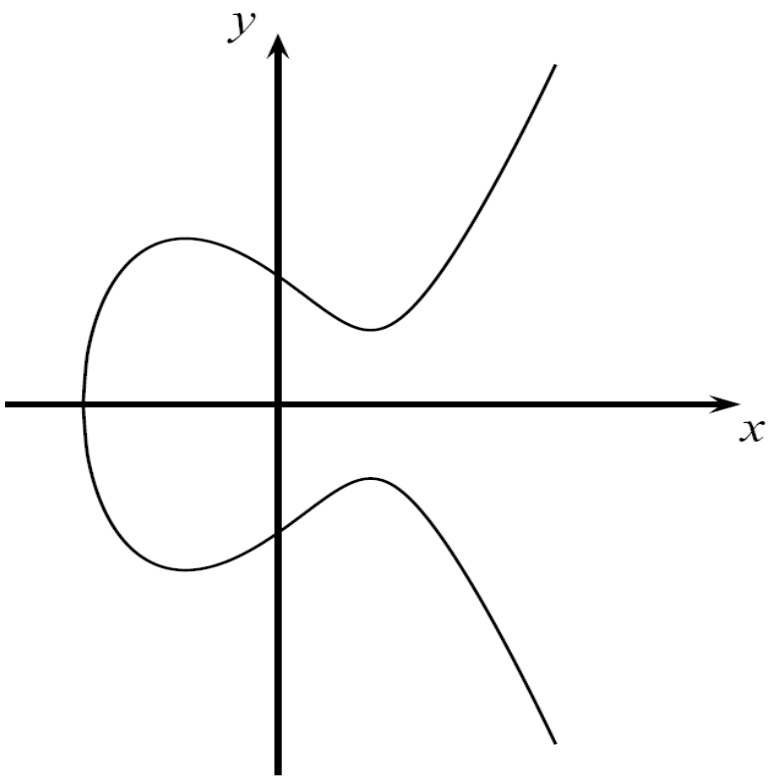
- Elliptic curves are polynomials that define points based on the equation:

$$y^2 = x^3 + ax + b$$

for parameters a,b that specify the exact shape of the curve.

- On the real numbers and with parameters $a, b \in \mathbb{R} - \{0\}$, an elliptic curve looks like this
- Elliptic curves can not just be defined over the real numbers *R* but over many other types of finite fields.

Example:
$y^2 = x^3 - 3x + 3$ over R



Elliptic Curves over finite fields

- In Informatics, we are interested in elliptic curves modulo prime *p*:

Definition: Elliptic Curves over prime fields

The elliptic curve over Z_p, p>3 is the set of all pairs $(x, y) \in Z_p$ *which fulfill*

$$y^2 = x^3 + ax + b \pmod{p}$$

together with an imaginary point of infinity θ.

where $a, b \in Z_p$ *and the condition* $4a^3 + 27b^2 \neq 0 \pmod{p}$.

- Note that $Z_p = \{0,1,\ldots, p-1\}$ is a set of integers with modulo p arithmetic

Elliptic Curve on a finite set of Integers

- Consider $y^2 = x^3 + 2x + 3 \pmod 5$

  $x = 0 \Rightarrow y^2 = 3 \Rightarrow$ no solution $\pmod 5$

  $x = 1 \Rightarrow y^2 = 6 = 1 \Rightarrow y = 1,4 \pmod 5$

  $x = 2 \Rightarrow y^2 = 15 = 0 \Rightarrow y = 0 \pmod 5$

  $x = 3 \Rightarrow y^2 = 36 = 1 \Rightarrow y = 1,4 \pmod 5$

  $x = 4 \Rightarrow y^2 = 75 = 0 \Rightarrow y = 0 \pmod 5$

- Then points on the elliptic curve are

  (1,1) (1,4) (2,0) (3,1) (3,4) (4,0) and the point at infinity: $\boldsymbol{\theta}$

Worked Example … Contd.

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Elliptic Curves: y^2 = x^3 + a*x + b
Enter the GF mod value: 13
Enter the coefficient a: 1
Enter the coefficient b: 1

Values in Z(13):

[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]

Points that satisfies the equation: y^2 = x^3 + 1*x + 1

(0, 12)
(0, 1)
(1, 9)
(1, 4)
(4, 11)
(4, 2)
(5, 12)
(5, 1)
(7, 0)
(7, 0)
(8, 12)
(8, 1)
(10, 7)
(10, 6)
(11, 11)
(11, 2)
(12, 8)
(12, 5)
```

Computations on Elliptic Curves

Generating a *group of points* on elliptic curves

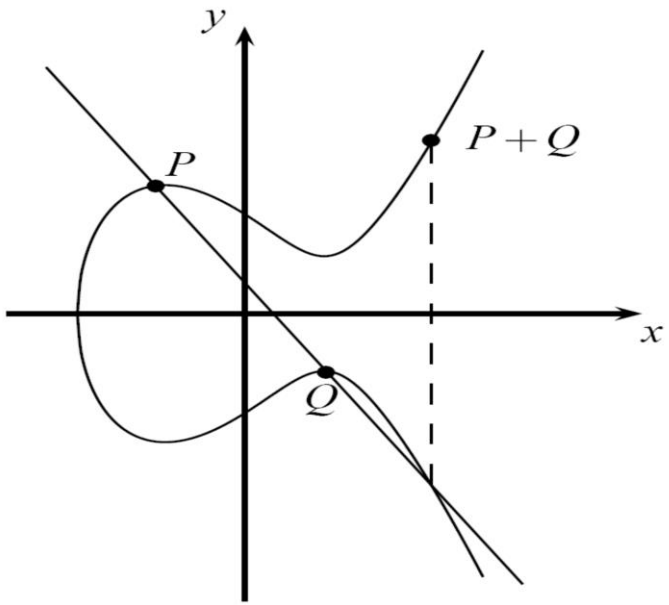
$$y^2 = x^3 + ax + b \pmod{p}$$

based on point addition operation *P+Q = R, i.e.,*

$$(x_1, y_1) + (x_2, y_2) = (x_3, y_3)$$

Geometric Interpretation of point addition operation

- *Draw straight line through P and Q; if P=Q use tangent line instead*
- *Mirror third intersection point of drawn line with the elliptic curve along the x-axis*



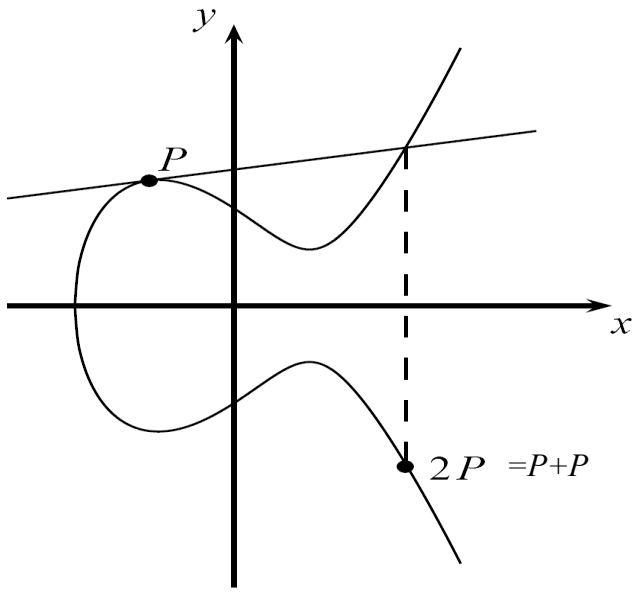
Point Addition

Elliptic Curve Point Addition and Doubling Formulas:

$x_3 = s^2 - x_1 - x_2 \mod p$ and $y_3 = s(x_1 - x_3) - y_1 \mod p$

where

$$s = \begin{cases} \dfrac{y_2 - y_1}{x_2 - x_1} \bmod p & \text{; if } P \neq Q \text{ (point addition)} \\ \dfrac{3x_1^2 + a}{2y_1} \bmod p & \text{; if } P = Q \text{ (point doubling)} \end{cases}$$



Point Doubling

Computations on Elliptic Curves (ctd.)

- **Example**: Given $E: y^2 = x^3+2x+2 \bmod 17$ and point $P=(5,1)$

Goal: Compute $2P = P+P = (5,1)+(5,1)= (x_3,y_3)$

$$s = \frac{3x_1^2 + a}{2y_1} = (2 \cdot 1)^{-1}(3 \cdot 5^2 + 2) = 2^{-1} \cdot 9 \equiv 9 \cdot 9 \equiv 13 \bmod 17$$

$$x_3 = s^2 - x_1 - x_2 = 13^2 - 5 - 5 = 159 \equiv 6 \bmod 17$$

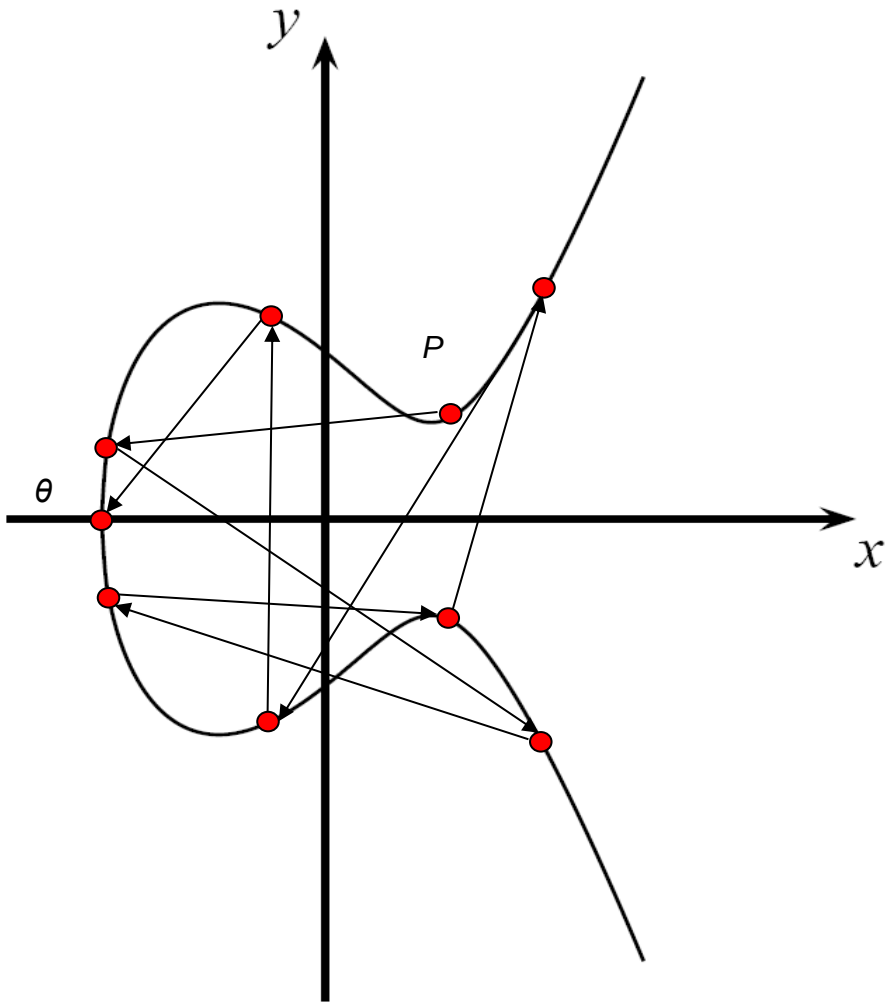
$$y_3 = s(x_1 - x_3) - y_1 = 13(5 - 6) - 1 = -14 \equiv 3 \bmod 17$$

Finally $2P = (5,1) + (5,1) = (6,3)$

Computations on Elliptic Curves (ctd.)

The points on an elliptic curve $E: y^2 = x^3+2x+2 \bmod 17$ and the point at infinity θ form cyclic subgroups

$2P = (5,1)+(5,1) = (6,3)$
$3P = 2P+P = (10,6)$
$4P = (3,1)$
$5P = (9,16)$
$6P = (16,13)$
$7P = (0,6)$
$8P = (13,7)$
$9P = (7,6)$
$10P = (7,11)$
$11P = (13,10)$
$12P = (0,11)$
$13P = (16,4)$
$14P = (9,1)$
$15P = (3,16)$
$16P = (10,11)$
$17P = (6,14)$
$18P = (5,16)$
$19P = \theta$



This elliptic curve has order #E = |E| = 19 since it contains 19 points in its cyclic group.

Defining *mP*

- $2P = P + P$
- $3P = P + P + P = 2P + P$
- …
- $mP = P + P + \ldots + P = (m-1)P + P$
- No matter how big *m* is, there is an efficient (quick) way to calculate *m*P.

Worked Example

If $P=(x_p, y_p)$ then $-P=(x_p,-y_p)$.

Example: Compute $-P$ on the elliptic curve $E_{23}(1,1)$ if $P=(13,7)$.

$$
\begin{aligned}
-P &= (13, -7) \\
&= (13 \bmod 23, -7 \bmod 23) \\
&= (13, 16)
\end{aligned}
$$

Number of Points on an Elliptic Curve

How many points can be on an arbitrary elliptic curve?

- Consider previous example: $E: y^2 = x^3+2x+2 \bmod 17$ has 19 points
- However, determining the point count on elliptic curves in general is hard

- But Hasse's theorem bounds the number of points to a restricted interval

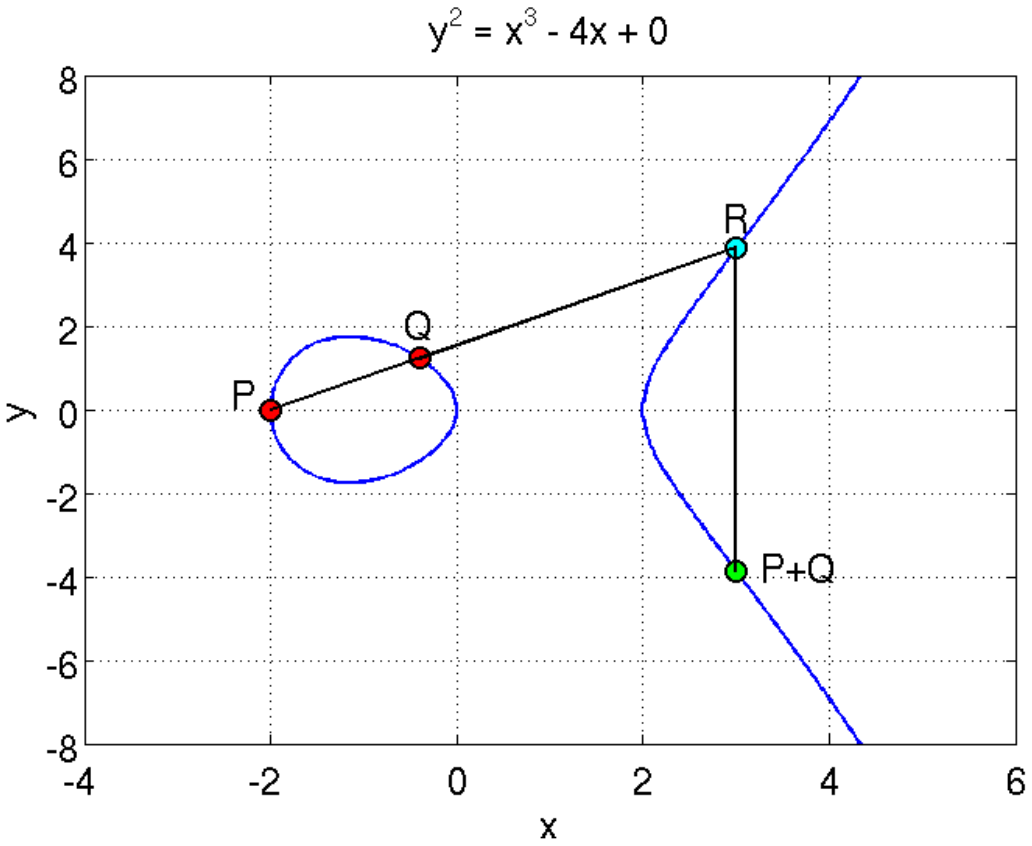
Definition: Hasse's Theorem:

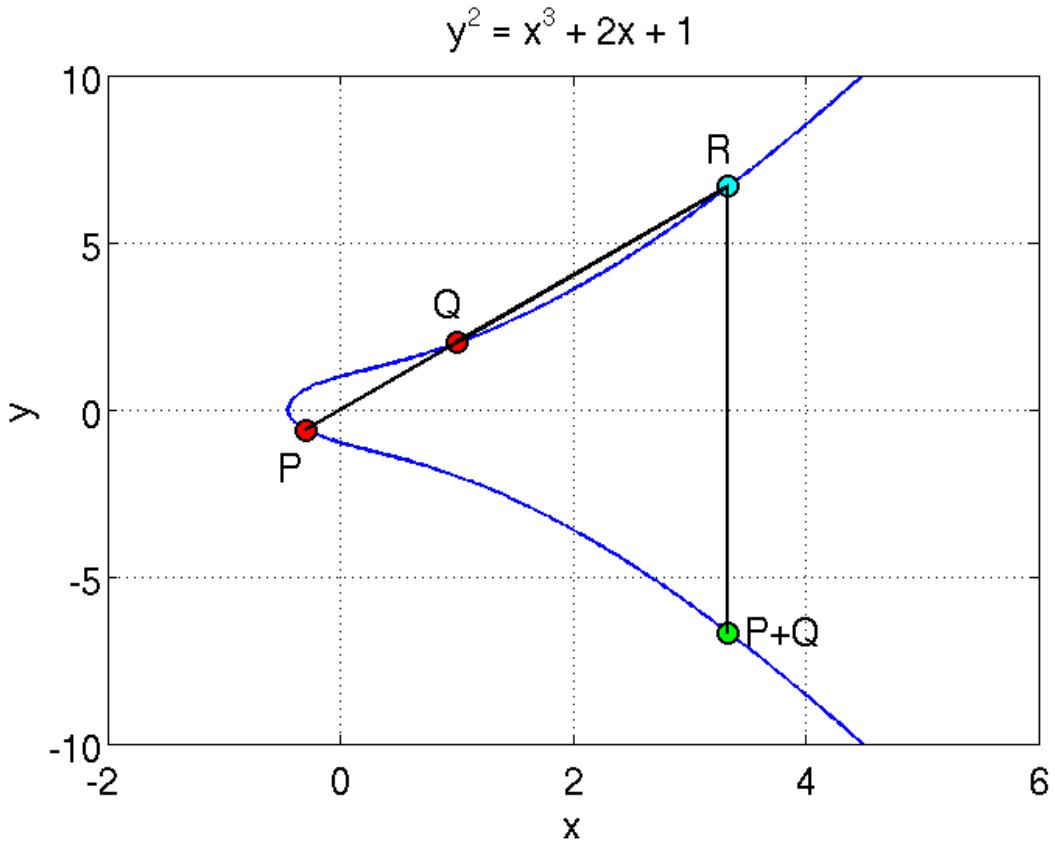
Given an elliptic curve module p, the number of points on the curve is denoted by #E and is bounded by

$$p+1-2\sqrt{p} \leq \#E \leq p+1+2\sqrt{p}$$

- **Interpretation:** The number of points is "close to" the prime p
- **Example:** To generate a curve with about 2^{160} points, a prime with a length of about 160 bits is required

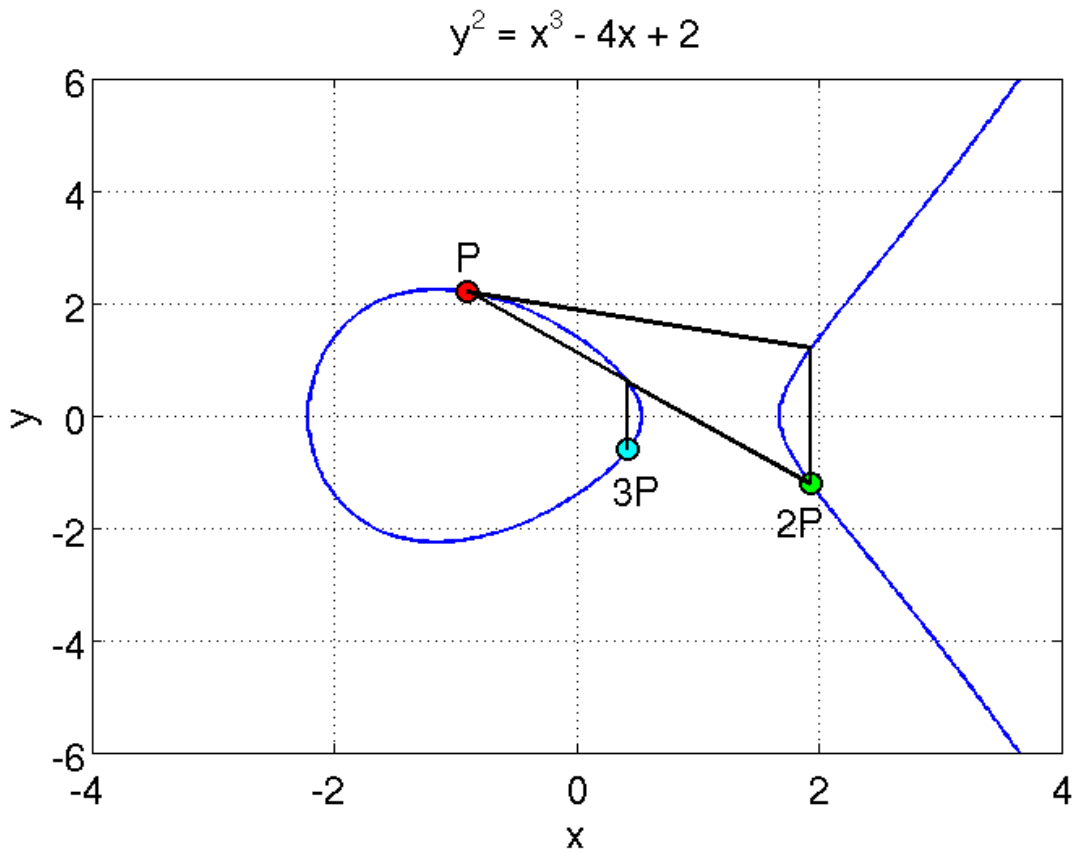
Graphical Representation

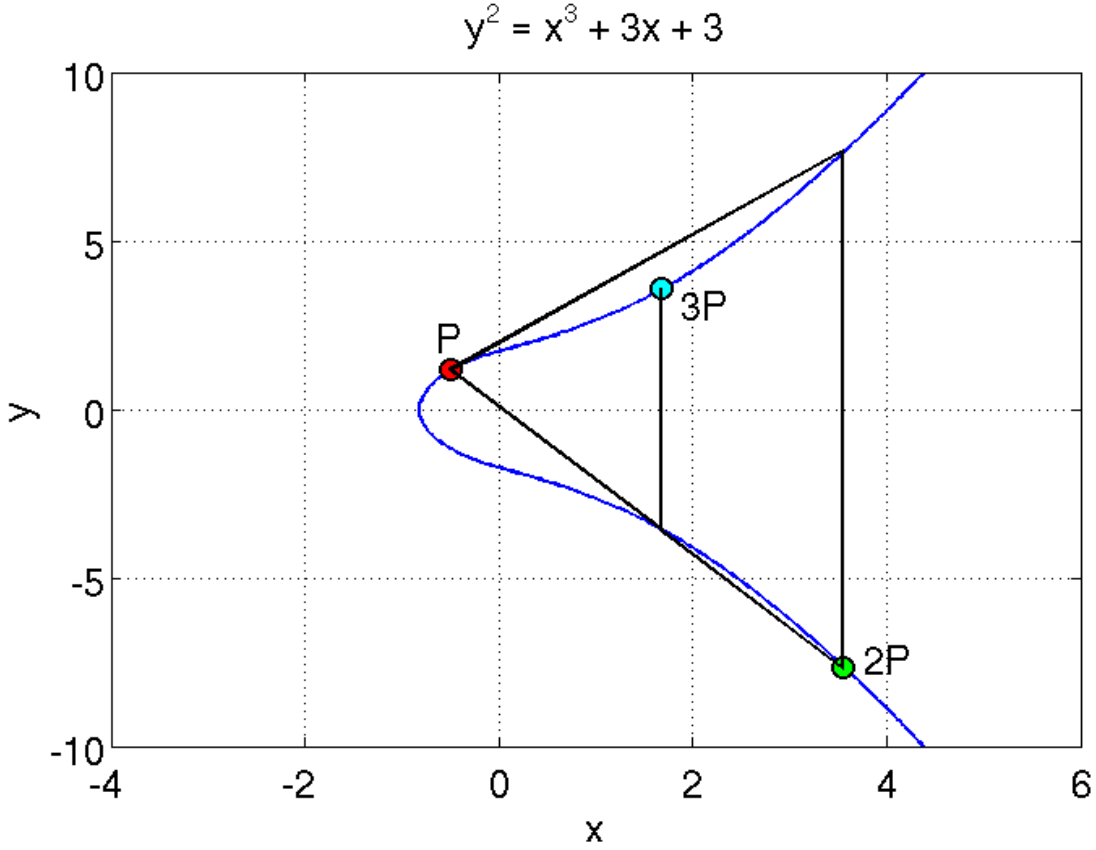




- Adding two points on the curve
- P and Q are added to obtain P+Q, which is a reflection of R along the X axis

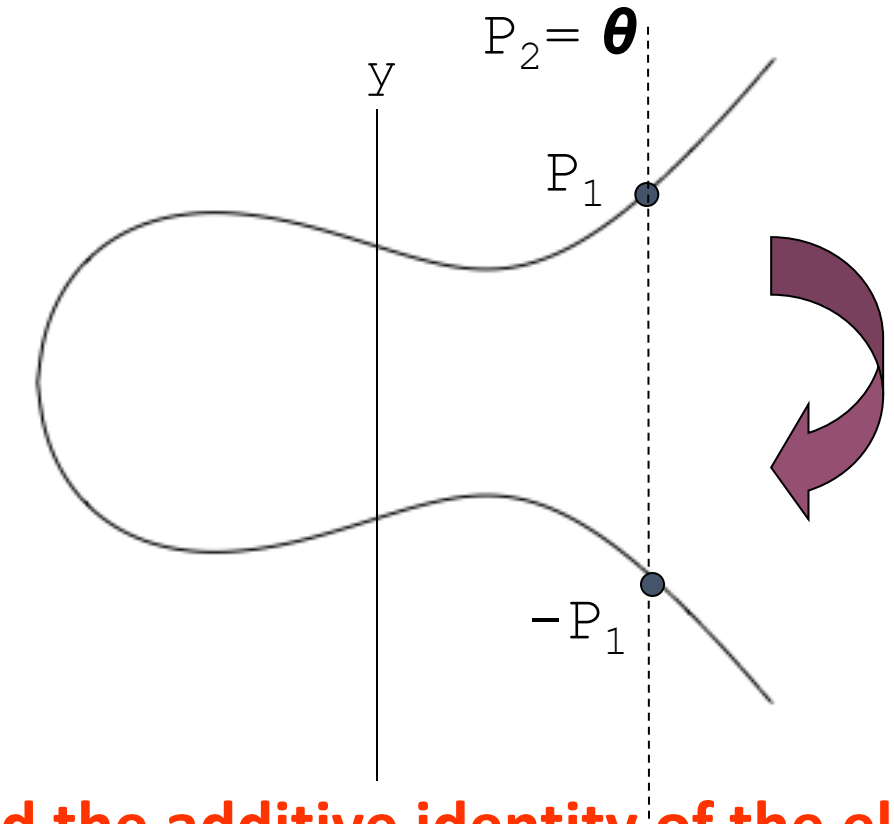
Graphical Representation





- A tangent at P is extended to cut the curve at a point; its reflection is 2P
- Adding P and 2P gives 3P
- Similarly, such operations can be performed as many times as desired to obtain **Q = kP**

Why do we need the reflection?



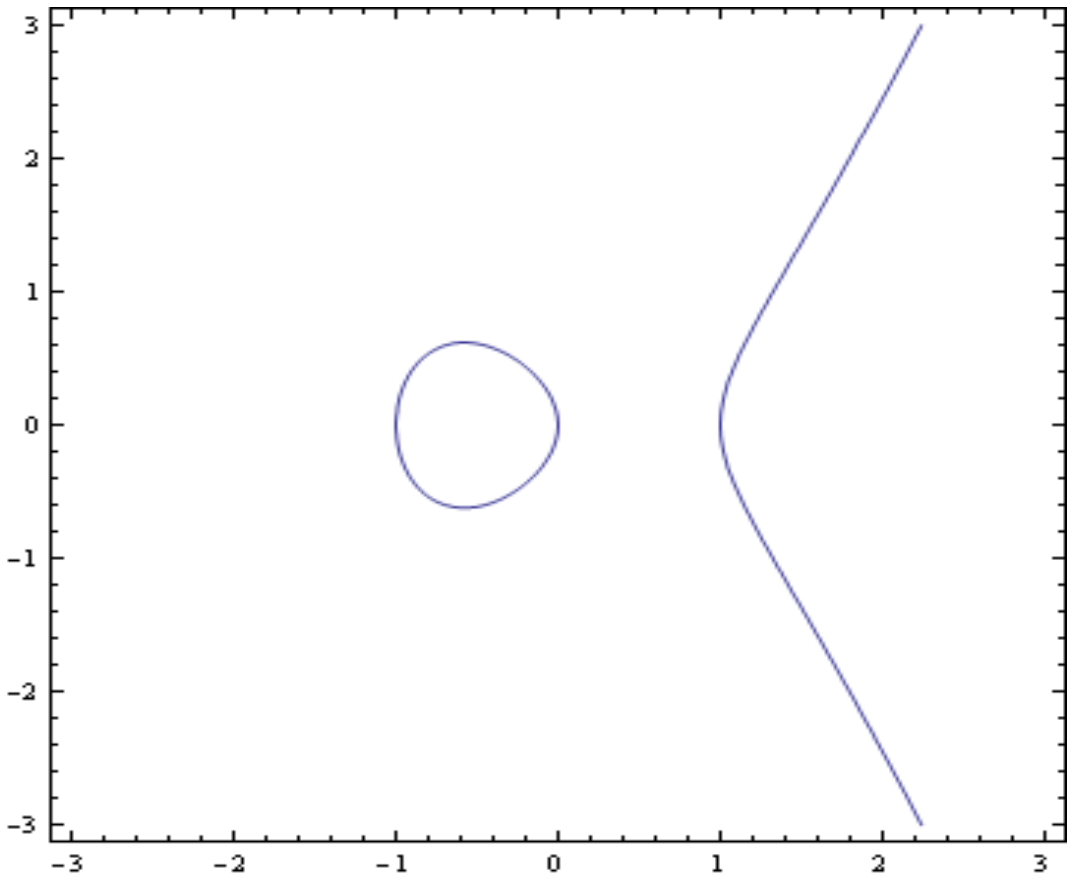
$$P_1 = P_1 + \theta = P_2$$

(or)

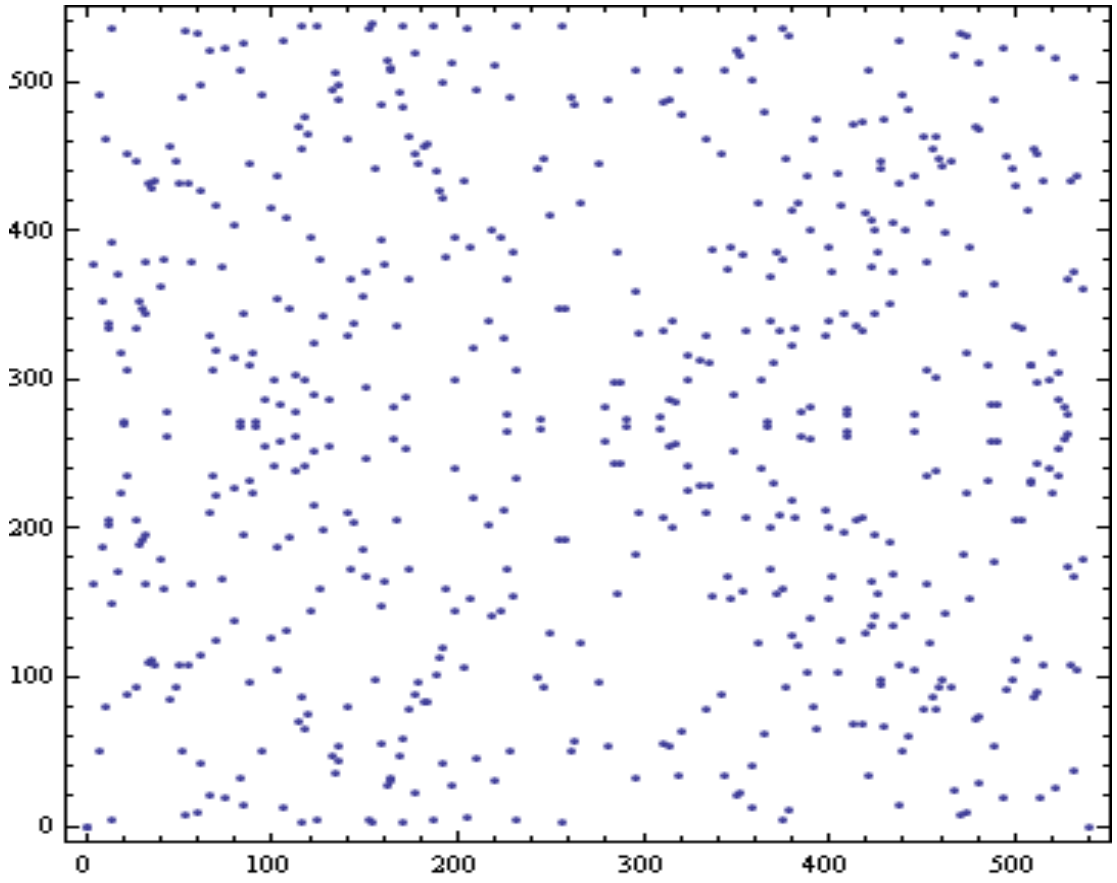
$$P_1 + (-P_1) = \theta$$

θ is called the additive identity of the elliptic curve group.

Elliptic Curve mod *p*

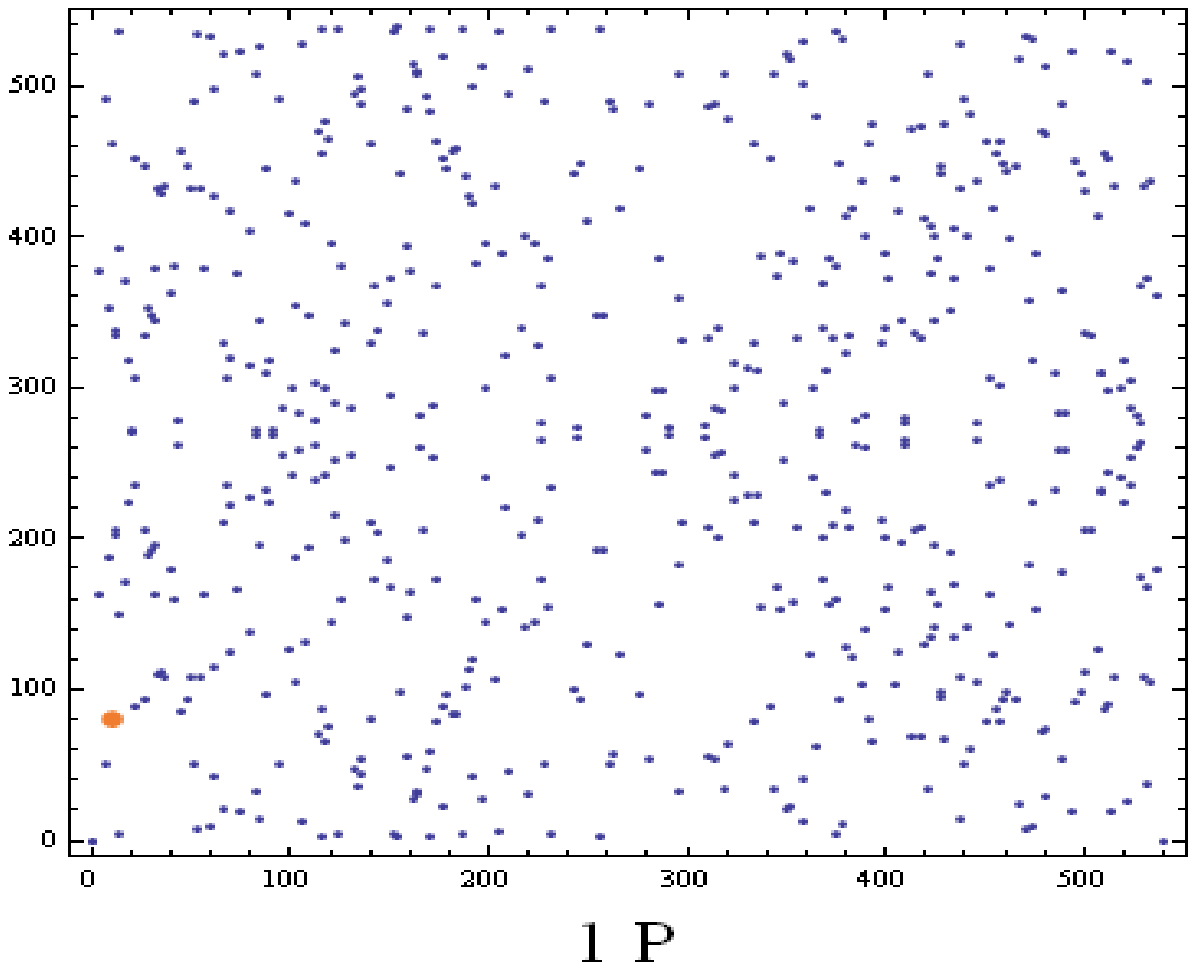


$y^2 = x^3 - x$

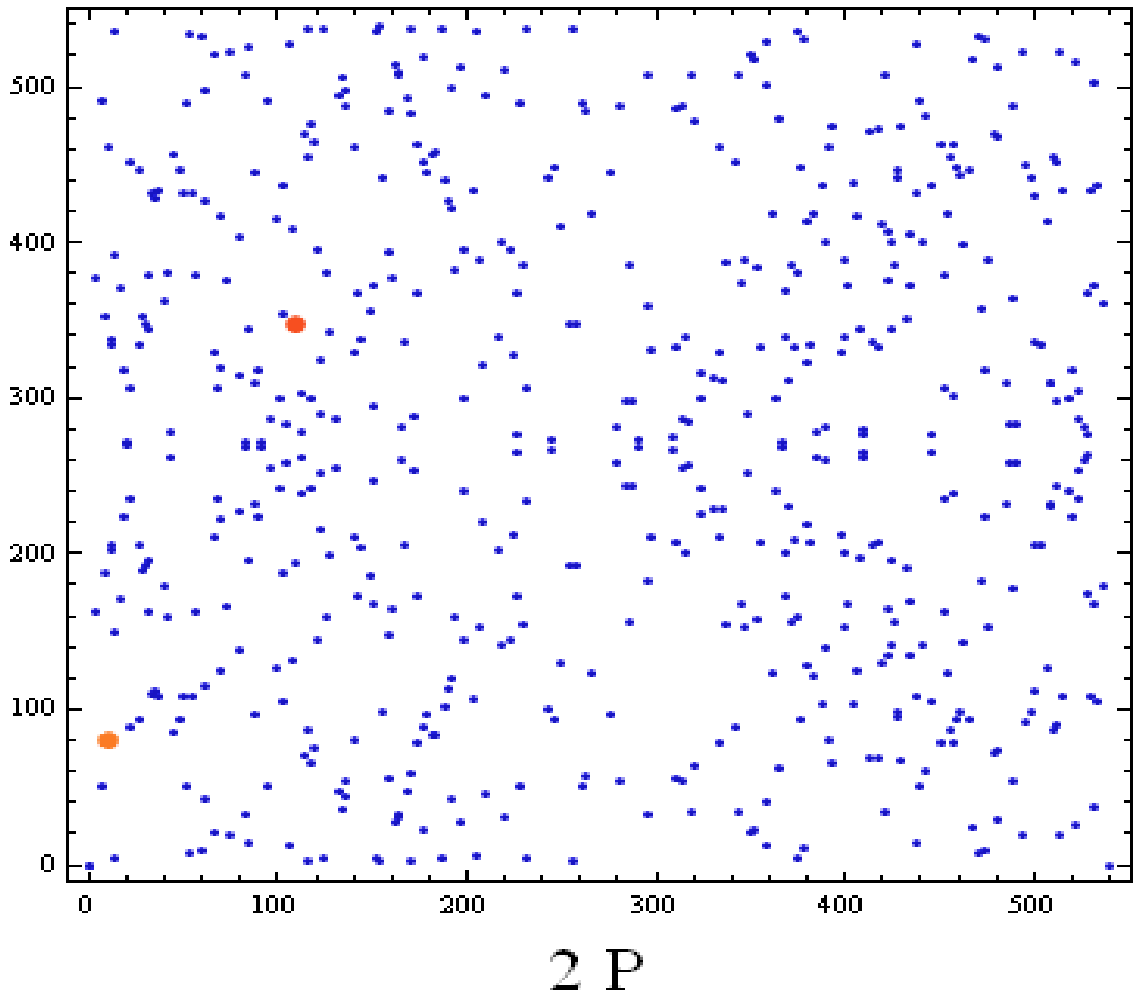


Elliptic Curve mod 541

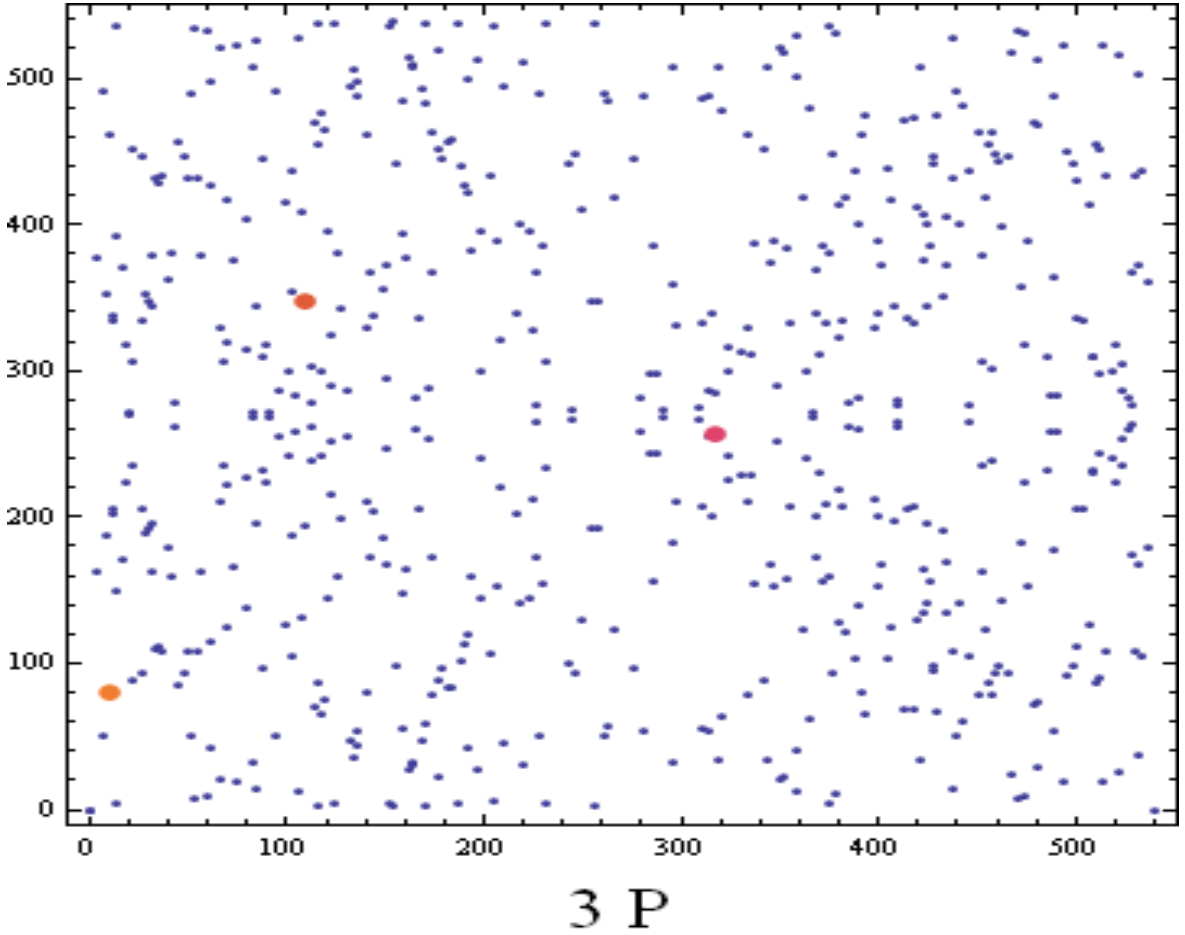
Example mod 541



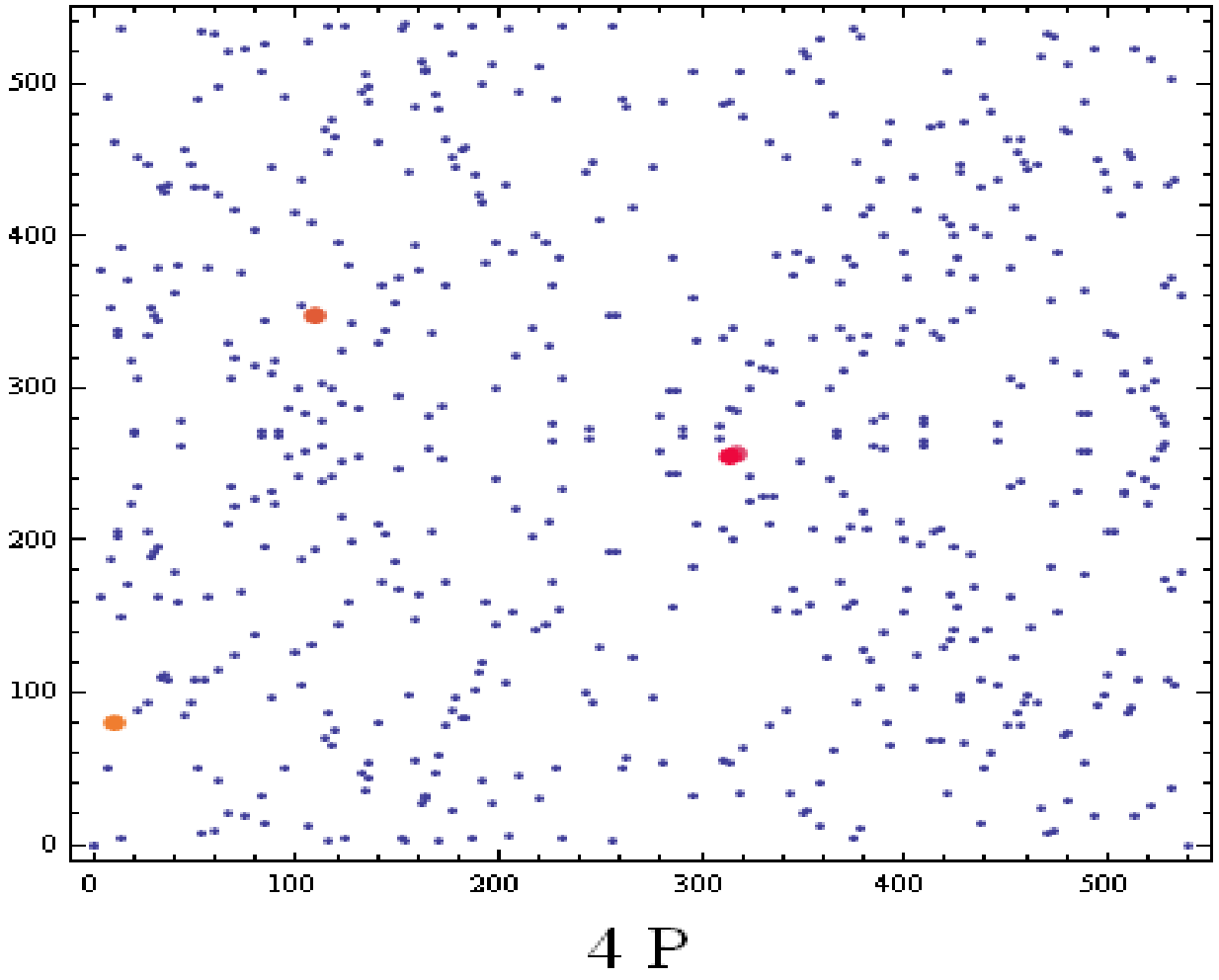
Example mod 541



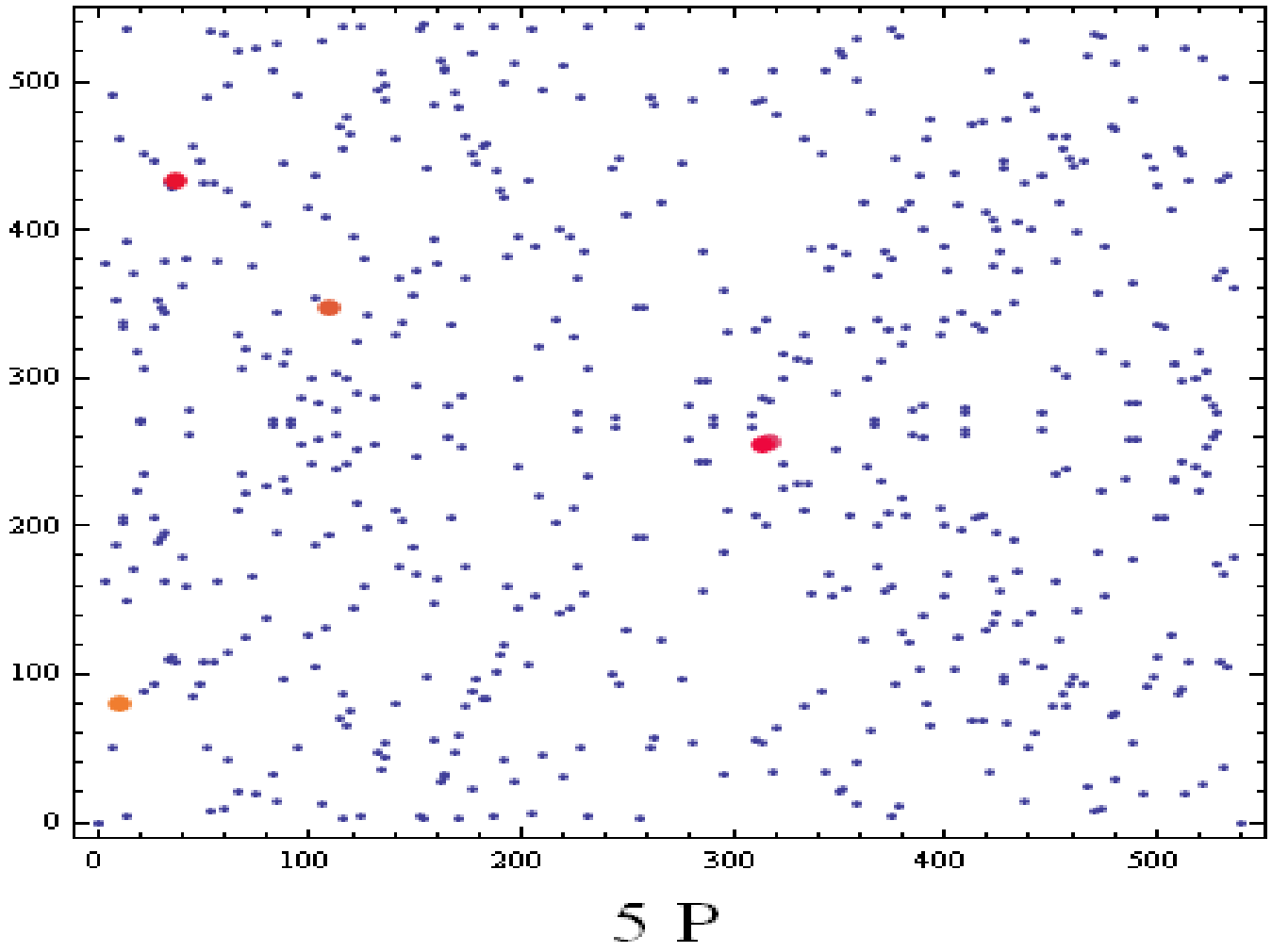
Example mod 541



Example mod 541

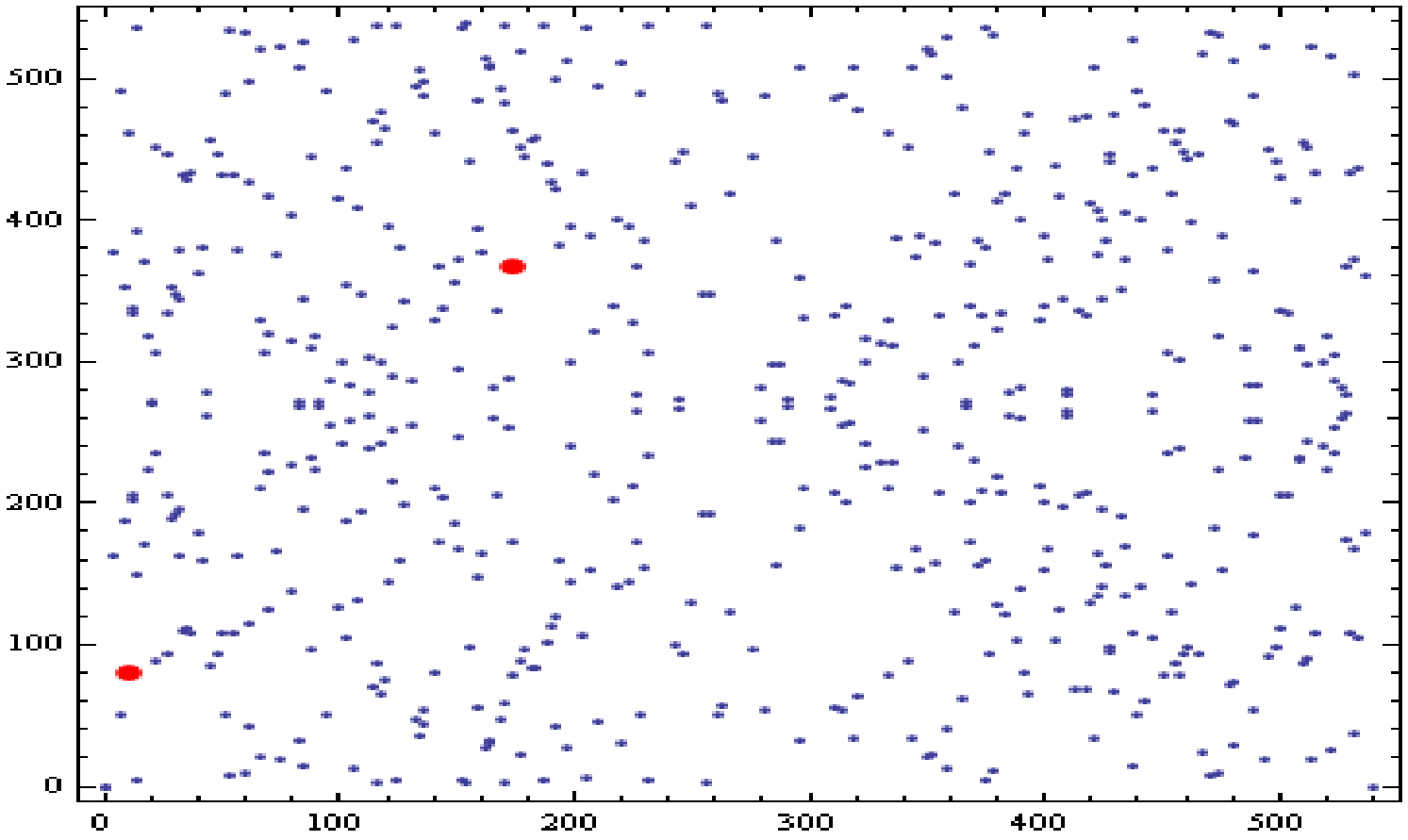


Example mod 541



Elliptic Curve Discrete Log Problem

What is m?



$$(173, 368) = m\,(10, 80)$$

Elliptic Curve Cryptography (ECC)

- ECC addition is analog of modulo multiply
- ECC repeated addition is analog of modulo exponentiation
- Need “hard” problem equiv to discrete log
  - Q=kP, where Q,P belong to a prime curve
  - is “easy” to compute Q given k,P
  - but “hard” to find k given Q,P
  - known as the elliptic curve logarithm problem

ECDLP

- Begin with an elliptic curve mod *p*, let P be a point, and let Q be a multiple of P.

The ECDLP is to find the value of *m* such that Q = *m*P.

- **We can simply calculate 2P, 3P, 4P, etc. But if *p* and *m* are large numbers, this could take trillions of years.**
- **Basically, we do not know a fast way to solve ECDLP.**

ECC Diffie-Hellman Key Exchange Protocol

Algorithm:

- Can do key exchange analogous to D-H
- Users select a suitable curve $E_q(a,b)$
- Select base point $G=(x_1,y_1)$
  - with large order n s.t. $nG=O$
- Alice & Bob select private keys $n_A<n$, $n_B<n$
- Compute public keys: $P_A=n_AG$, $P_B=n_BG$
- Compute shared key: $K=n_AP_B$, $K=n_BP_A$
  - same since $K=n_An_BG$
- **Attacker would need to find k, hard**

Example

- Let's use $y^2 = x^3 - x$ with $p = 541$, $B = (10, 80)$.
- Alice picks $p_A = 20$. Bob picks $p_B = 103$.
- $Q_A = 20\ (10, 80) = (519, 241)$.
- $Q_B = 103\ (10, 80) = (85, 345)$.
- When Alice gets Q_B, she finds $20Q_B = (353, 158)$.
- When Bob gets Q_A, he finds $103Q_A = (353, 158)$.
- They both use $K = 353$ for their key.

ECC Diffie-Hellman Key Exchange Protocol

Example: $p = 211$; for $E_p(0,-4)$, choose $G = (2,2)$. Note that $241G = O$. $n_A = 121$, and $P_A = 121(2,2) = (115,48)$. $n_B = 203$ and $P_B = 203(2,2) = (130,203)$. The shared secret key is then $121(130,203) = 203(115,48) = (161,169)$.

For choosing a single number as the secret key, we could simply use the x coordinates or some simple function of the x coordinate.

Is it secure?

- Eve knows the elliptic curve, the prime p, the original point B, and the points $Q_A = p_A B$ and $Q_B = p_B B$.
- To break, Eve needs to find p_A or p_B. To get either value, Eve needs to solve the ECDLP.
- **No one knows how to do this in a reasonable length of time.**

ECC Encryption/Decryption Protocol

A. Key Generation in ECC:

1.Select an elliptic curve equation:

$E : y^2 = x^3 + ax + b \mod p$, where p is a prime number, and a, b defines the curve.

2.Select a base point P on the curve. This point is shared publicly.

3.Generate private key d:

Randomly select a private key d∈[1, n−1], where n is the order of the elliptic curve.

1.Generate public key Q:

Compute the public key Q=d·P mod p (scalar multiplication).

ECC Encryption/Decryption Protocol

B. Encryption Algorithm:

Steps:

1.Represent the message as a point M on the elliptic curve.

2.Select a random integer k∈[1,n−1]. (used for one-time encryption).

3.Calculate two points:

$C_1 = k \cdot P$, where Q is the recipient's(Bob) public key.

$C_2 = M + k \cdot Q$

4.The ciphertext C is the pair of points (C_1, C_2).

ECC Encryption/Decryption Protocol

ECC Decryption Algorithm:

Let:

Ciphertext C $= (\boldsymbol{C_1}, \boldsymbol{C_2})$ be the two points received from the sender(Alice).

Private key d of the recipient (Bob).

Steps:

1. Compute $d.C_1$, where d is the recipient's private key.
2. Decrypt the message: $M = C_2 - d \cdot C_1$

ECC Encryption/Decryption … Contd.

Example: Take $p = 751$, $E_p(-1,188)$ and $G = (0,376)$. Assume that $P_m = (562,201)$ is to be sent and that the sender chooses a random number $k = 386$. Assume that the receiver's public key is $P_B = (201,5)$. We have $386(0,376) = (676,558)$, and $(562,201) + 386(201,5) = (385,328)$. Consequently, $\{(676,558), (385,328)\}$ is sent as the ciphertext.

Practice Questions

1) Consider the elliptic curve $E_7(2,1)$; that is the curve is defined by $y^2=x^3+2x+1$ with a modulus of p=7. Determine all the points in $E_7(2,1)$.

2) What are the negatives of the following elliptic curve points over Z_{17}?

   a) P=(5,8) b) Q=(3,0) c) R=(0,6)

3)For $E_{11}(1,7)$, consider the point G=(3,2). Compute the multiple of G from 2G through 13G.

4)Assume a secure communication happens between User A and B using elliptic curve cryptosystem. The cryptosystem parameters are $E_{11}(1, 6)$ and G=(2,7). B's private key is $n_B=7$.

   a) Find B's public key P_B.

   b) A wishes to encrypt the message $P_m=(10, 9)$ and chooses a random value k=3. Determine the ciphertext C_m.

   c) Show the calculation by which B recovers P_m from C_m.

Comparable Key Sizes for Equivalent Security

Symmetric Encryption (Key Size in bits)	RSA and Diffie-Hellman (modulus size in bits)	ECC Key Size in bits
56	512	112
80	1024	160
112	2048	224
128	3072	256
192	7680	384
256	15360	512